

QuickChat Messaging Application

Overview

QuickChat is a simple Java-based messaging application that allows users to register, log in, send messages, and view received messages in an inbox. The project demonstrates the use of object-oriented programming principles such as classes, objects, encapsulation, collections, and validation. The application also includes JUnit 5 unit tests to verify that all major features work correctly.

Features

- User registration with validation
- Username, password, and phone number checking
- User login and logout
- Send messages between users
- Store and display sent messages
- Inbox system for receiving messages
- Message ID and message hash generation
- JUnit 5 testing for all major classes

Technologies Used

- Java
- NetBeans IDE
- JUnit 5
- Java Collections Framework (HashMap, ArrayList)
- Regular Expressions (Pattern)

Project Structure

QuickChat/

```
|
|└─ Source Packages
|  |└─ Login.java
|  |└─ User.java
|  |└─ Message.java
```

```
| └─ MessagingApp.java
|
| └─ Test Packages
|   └─ LoginTest.java
|   └─ UserTest.java
|   └─ MessageTest.java
|   └─ MessagingAppTest.java
|
└─ README.md
```

Class Descriptions

Login Class

Handles user registration and login validation. It checks username formatting, password complexity, and South African phone number formatting before storing user details.

User Class

Represents a user account in the system. It stores login information and manages the user inbox where received messages are saved.

Message Class

Handles message creation, validation, hashing, storing, and displaying. It also tracks the total number of sent messages.

MessagingApp Class

Acts as the main controller of the application. It manages users, login sessions, messaging functionality, and the menu system.

Validation Rules

Username Rules

- Must contain an underscore _
- Must follow the required length restrictions

Password Rules

Password must contain:

- At least 8 characters

- One uppercase letter
- One number
- One special character

Phone Number Rules

- Must use South African international format
- +27831234567

JUnit Testing

The application includes unit tests for:

- Login validation
- Password checking
- Message validation
- Sending messages
- User inbox functionality
- Registration and login systems

The tests ensure that all methods behave correctly and help detect errors early during development.

Example Login Details

Username: jo_hn

Password: Password1!

Phone: +27831234567

Future Improvements

- Graphical User Interface (GUI)
- Database integration
- Message encryption
- Delete messages feature
- Search messages feature
- Real-time messaging
- File attachments

Author

Developed as a Java Object-Oriented Programming and Unit Testing project using NetBeans and JUnit 5.

License

This project is for educational purposes.